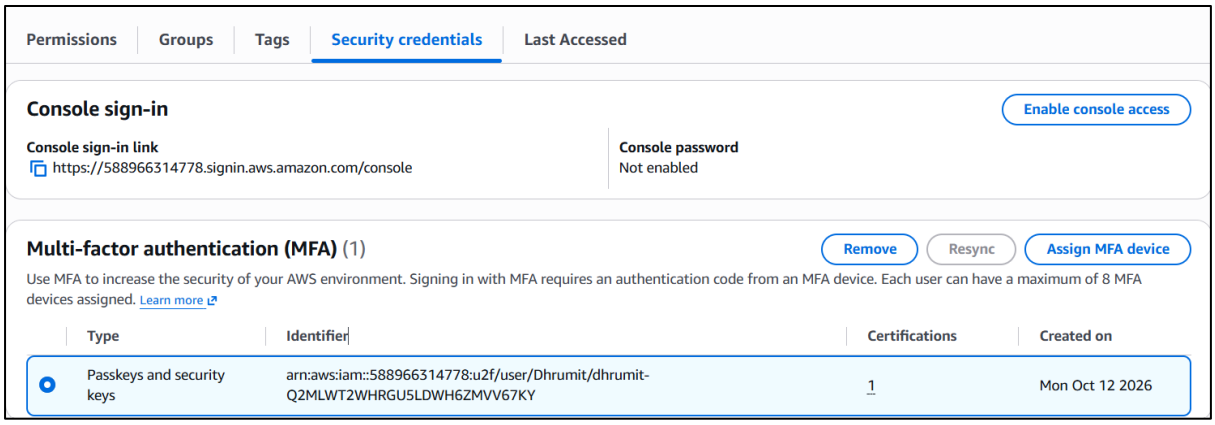
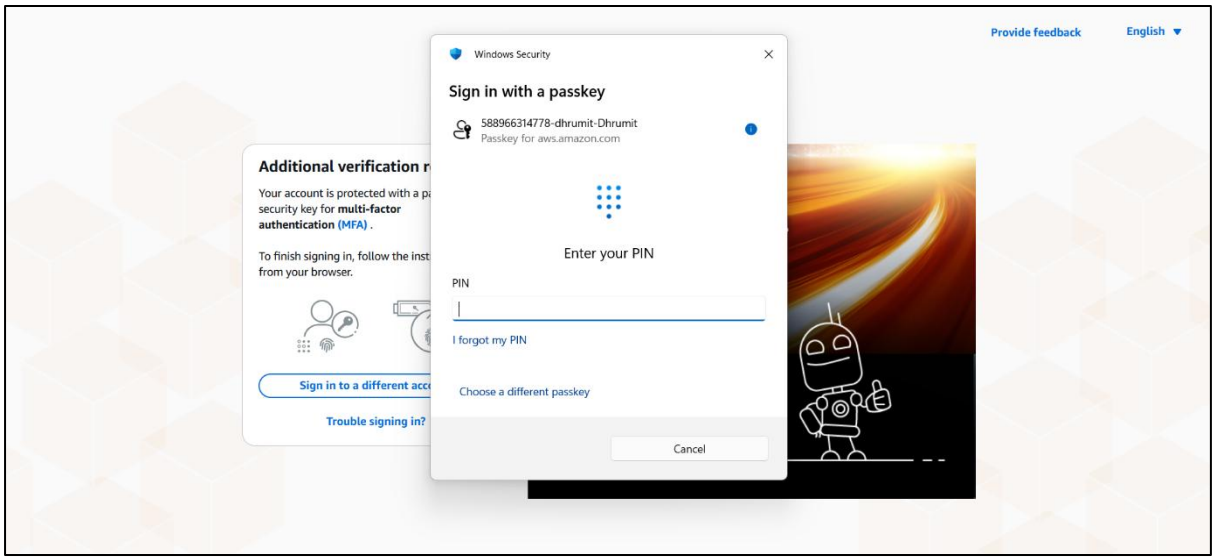
Security In The Cloud – Assignment-3

1. Enable Multi-Factor Authentication (MFA) on your personal AWS or Azure cloud account using the built-in authenticator app option, and take a screenshot of the MFA setup confirmation page.



2. Simulate a login to your cloud account from a new browser or incognito mode and document the steps and prompts you receive with MFA enabled.



3. Disable MFA temporarily on your cloud account, attempt to log in again, and note the difference in the login experience and security prompts .Hint: Compare the number of steps and any warnings shown with and without MFA4

Security In The Cloud – Assignment-3

Feature	With MFA enabled	Without MFA
Login steps	Password and an additional verification step	Usually username/account details and password
Security verification	MFA code or security key may be required	No MFA verification step
Account protection	Stronger protection against stolen passwords	More vulnerable if the password is compromised
Login experience	Takes a little longer	Usually faster
Security warnings	May show MFA setup or verification prompts	AWS may recommend enabling MFA

4. Write a short scenario (3-5 lines) describing how an attacker might try to access a cloud account without MFA, and explain how enabling MFA would block or alert you to this attempt.

Scenario: Cloud Account Attack Without MFA

1. An attacker obtains a user's AWS password through a phishing email.
2. Without MFA, the attacker may be able to log in using only the stolen password.
3. With MFA enabled, the attacker also needs a second verification factor, such as an authenticator code.
4. The attacker is less likely to gain access, even if the password is compromised.
5. AWS security monitoring and alerts can help identify suspicious login attempts.